

Causal Inference and ML: Simpson's Paradox in Drug Recovery

Analyze the effect of a drug on recovery and demonstrate Simpson's paradox caused by confounding variables

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Github Repository:

<https://github.com/polinato/simpsons-paradox-drug-recovery>

Simpson's Paradox in Drug Recovery

Dataset

800 patients

Treatment: drug (1/0)

Outcome: recover (1/0)

Confounders: age,
SES, hospital_rating



Naive Model not causal

The drug appears to increase recovery rates by about 10%

OLS Regression Results

Dep. Variable:	recover	R-squared:	0.010			
Model:	OLS	Adj. R-squared:	0.008			
Method:	Least Squares	F-statistic:	7.787			
Date:	Sun, 31 May 2026	Prob (F-statistic):	0.00539			
Time:	13:31:00	Log-Likelihood:	-570.19			
No. Observations:	800	AIC:	1144.			
Df Residuals:	798	BIC:	1154.			
Df Model:	1					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

Intercept	0.5150	0.025	20.844	0.000	0.467	0.563
drug[T.Yes]	0.0975	0.035	2.790	0.005	0.029	0.166
=====						
Omnibus:	3313.675	Durbin-Watson:	2.155			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	128.405			
Skew:	-0.253	Prob(JB):	1.31e-28			
Kurtosis:	1.104	Cond. No.	2.62			
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Simpson's Paradox in Drug Recovery

Cause

Treatment imbalance

- Drug is mostly given to Old patients
- Old patients have higher recovery rates
- The drug group looks artificially better in total

drug	No	Yes
age		
old	100	300
Young	300	100

age	drug	
Old	No	0.780000
	Yes	0.716667
Young	No	0.426667
	Yes	0.300000

Within each group, the drug reduces recovery

Controlled model

After controlling for age, SES, and hospital rating, the estimated effect of the drug becomes negative

OLS Regression Results

Dep. Variable:	recover	R-squared:	0.126			
Model:	OLS	Adj. R-squared:	0.122			
Method:	Least Squares	F-statistic:	28.63			
Date:	Sun, 31 May 2026	Prob (F-statistic):	3.01e-22			
Time:	13:31:19	Log-Likelihood:	-520.25			
No. Observations:	800	AIC:	1051.			
Df Residuals:	795	BIC:	1074.			
Df Model:	4					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

Intercept	0.7385	0.286	2.584	0.010	0.178	1.299
drug[T.Yes]	-0.0746	0.040	-1.875	0.061	-0.153	0.004
SES[T.Low]	0.0602	0.035	1.713	0.087	-0.009	0.129
age[T.Young]	-0.3827	0.041	-9.236	0.000	-0.464	-0.301
hospital_rating	0.0003	0.003	0.099	0.921	-0.006	0.007
=====						
Omnibus:	14550.035	Durbin-Watson:	2.196			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	70.838			
Skew:	-0.211	Prob(JB):	4.15e-16			
Kurtosis:	1.605	Cond. No.	1.44e+03			
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